

SMARTBRIDGE	Date: 15 March 2026	Team ID: 6aba389eb2817f097463f0f8	Max Marks: 2 Marks
Project Name: LegalEase		Product: SKILL WALLET (A SmartBridge Product)	

TECHNOLOGY STACK

Define Your Technology Stack: LegalEase - AI-Powered Legal Document Generator

LegalEase leverages state-of-the-art Generative AI to simplify the creation of legal documents by providing customizable, accurate, and editable templates for a wide range of use cases. Users can generate employment contracts, lease agreements, NDAs, and more—all tailored to their specific inputs like involved parties, effective dates, and key terms.

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit UI	Provides an intuitive and responsive web interface for user inputs (document type, parties, terms, dates), editable text previews, and seamless multi-format exports (.TXT, .DOCX, .PDF).
2	Backend / Server-Side	FastAPI & Uvicorn	Handles API requests efficiently using high-performance asynchronous execution, processing JSON payloads, and managing document generation routes via Pydantic models.
3	AI Core / Model Layer	Google Gemini 1.5 Pro (Google Generative AI SDK)	Utilizes advanced NLP with long context window support to understand legal jargon, enforce structured formatting, and draft comprehensive legal clauses dynamically.
4	Document Formatting & Export	python-docx, FPDF, Pillow	Converts AI-generated text into formatted Microsoft Word (.docx) files and branded PDF documents with embedded logos, custom headers, footers, and structured tables.
5	Environment & Dependencies	Python 3.10+, python-dotenv	Ensures secure API key management via environment variables (.env) and reliable dependency handling across local and cloud environments.
6	Deployment & Server	Render / Streamlit Community Cloud / VPS	Enables scalable, production-ready hosting for both backend REST API endpoints and frontend web applications.

